

Course name: Openshift Container Platform

Pre-requisites:

- Knowledge of basic Unix / Linux commands will be good to have

Docker Introduction

- What is Docker?
- Why Docker?
- Docker Architecture
- Understanding the Docker components
- Docker Terminologies
- Docker Internals

Docker Installation

- Docker Editions
- Hardware and Software Requirements
- Installing Docker
- Docker Config Files and Settings
- Lab: Installing Docker on Linux

Managing Container Lifecycle

- Creating, Starting and Listing Containers
- Attaching and Detaching from Container
- Fetching Container details using “inspect”
- Executing Commands inside a Container
- Pause, Stop, Start, Restart and Kill a Container
- Container Metrics via stats
- Limiting Memory and CPU utilization
- Storage Utilization of a Container
- Expose Container Application using Host Port
- Lab: Managing Container Lifecycle

Working with Docker Images

- Searching, Filtering and Pulling Images
- Building Custom Images using Dockerfile
- Understanding Image Layers
- Publishing the Custom Image

- Lab: Working with Docker Images Lab: Building Custom Images Lab: Working with Docker Hub
- Lab: Setting up Local Registry Service

Docker Volume Management

- Understanding need of Volume service in Docker
- Understanding various Volume Plug-ins
- Create and Manage Volumes
- Start a container with a Volume
- Populate data in a Volume using a Container
- Use a read-only Volume
- Migrating Volumes between Containers
- Lab: Creating and Manage volumes

Docker Networking

- Understanding various Network Plug-ins
- Understanding the default Docker Networking
- Working with Bridge Network
- Working with Host and None Networks
- Lab: Managing Docker Networking

Course Overview

Kubernetes & Openshift

Kubernetes Architecture

- Kubernetes Terminology
- Kubernetes Components
- Kubernetes Cluster Architecture
- Understanding Kubernetes Master Components
 - Kube-apiserver
 - ETCD
 - Kube-scheduler
 - Kube-controller
 - Kube-DNS
- Understanding Kubernetes Node Components
 - Kube-proxy
 - Kubelet
 - Container Runtime

Introduction to OpenShift

- What is OpenShift?
- OpenShift Features
- OpenShift Architecture
- Difference Between Various OpenShift Editions
- Hosted vs Self-Managed OpenShift
- What is OpenShift Control Plane?
- Understanding Cluster Operators

Installing and Connecting to an OpenShift Cluster

- OpenShift installation step from scratch
- UPI vs IPI Installation Methods
- Connecting to a Cluster
- Troubleshooting a Cluster
- Lab: Deploying OpenShift Cluster

OpenShift Web Console

- Cluster Administration with the Web Console
- Managing Workloads with the Web Console
- Monitoring OpenShift with the Web Console
- Lab: Accessing OpenShift dashboard
- Lab: Deploying managing Applications using the dashboard

Course Overview

Working with Pod

- Pod Overview
- Understanding Pod Lifecycle
- Multi-container Pod
- Static Pod
- Init Containers
- Labels, Selectors & Annotations
- Lab: Imperative Commands and Formatting Output with kubectl
- Lab: Working with Single Container Pods
- Lab: Creating multi container Pod

Openshift Networking and Service

- Cluster Communications
- Pod and Node Networking
- Container Network Interface (CNI)
- Service Networking: ClusterIP, NodePort
- Network Policies
- Lab: Exposing Applications using various types of Services
- Lab: Create Network Policies to control traffic flow

Application Lifecycle Management

- Pods: Single Container, Multi Container, Static, Init
- Deploying Applications in the Kubernetes Cluster
- Controllers: RS, Deployment, and DaemonSets
- Deploying an Application, Rolling Updates, and Rollbacks
- Configuring an Application for High Availability and Scale
- Imperative Commands & Manifests (YML Intro)
- Lab: Deploying Application using Replica Set
- Lab: Rolling Updates and Rollbacks using Deployment
- Lab: Deploying Application using Daemon Set

Working with Kubernetes Scheduler

- Pod Scheduling within the Kubernetes Cluster
- Configuring the Kubernetes Scheduler
- Running Multiple Schedulers for Multiple Pods
- Taints, Tolerances, Node Selector, labels & Selectors
- Scheduling Pods with Resource Limits and Label Selectors
- Displaying Scheduler Events
- Lab: Manually scheduling Pod
- Lab: Scheduling Pod based on Node Selector and Labels
- Lab: Taints and Tolerations

Course Overview

Storage

- Managing Data in the Kubernetes Cluster
- EmptyDir, hostPath, PV, PVC, StorageClass
- Volume Access Modes
- Applications with Persistent Storage
- ConfigMaps, Secrets
- Lab: Working with Kubernetes Volume Service
- Lab: Configuring Dynamic Volume Service
- Lab: Working with ConfigMaps and Secrets

Logging and Monitoring

- Describe Resources
- Pod/container logs
- Metric Server & top command
- Events
- Lab: Working on Logs and Events
- Lab: Working with Metric Server
- Lab: working with HPA

Managing Cluster

- Managing the Kubernetes Cluster
- Upgrading the Kubernetes Cluster
- Backing Up and Restoring a Kubernetes Cluster
- Lab: Upgrading Kubernetes Cluster

Troubleshooting

- Control Plane Failures
- Node Failure
- Application Failure
- Components manifests and errors
- Scenarios and Solutions
- Lab: Troubleshooting in Kubernetes

Configuring Security for Users and Applications

- Authentication and Authorization
- What is the Authentication Operator?
- Different Identity Providers
- Using HTPasswd Identity Provider
- Adding Users and Assigning Permissions
- Difference between Rule, Role, and Binding
- Default Roles in OpenShift
- User Types in OpenShift

Course Overview

- Managing Secrets / Config Map in OpenShift
- Lab: Managing users and Groups in OpenShift
- Lab: Working with RBAC
- Lab: Working with Secrets and ConfigMaps in OpenShift

Helm :

Introduction to Helm

- Introduction to Helm
- Why Helm
- The Helm Ecosystem
- Charts
- Templates
- Default Templates
- Chart Repositories
- Hosting Your Own Repository
- Commercial Repositories
- Repository Requirements
- Using Repositories
- Using Helm

Charts and Templates

- Creating a Chart
- Chart Contents
- Chart.yaml
- Templates
- Template Structure
- A Simple Manifest
- Go Templates
- Add Go Template
- Getting Value from values.yaml
- Values.yaml
- Render the Template
- Override Values
- Template Syntax
- Default Functions
- Template Pipelines
- Template Conditionals